COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010 DATA STRUCTURE AND ALGORITHM	**SECTION:**	CPE21S4
DATE PERFORMED:	7/17/26	**DATE SUBMITTED:**	7/17/26
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ACTIVITY NO. 2 ARRAYS, POINTERS AND DYNAMIC MEMORY ALLOCATION

1. PROCEDURE OUTPUT

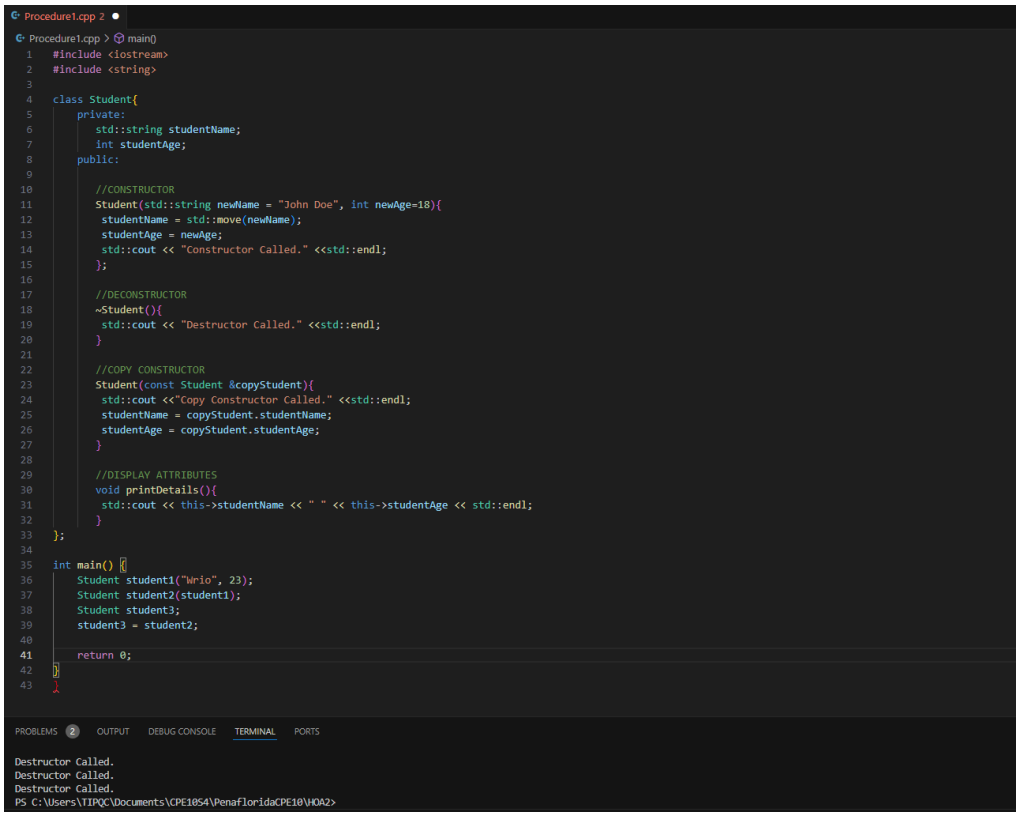
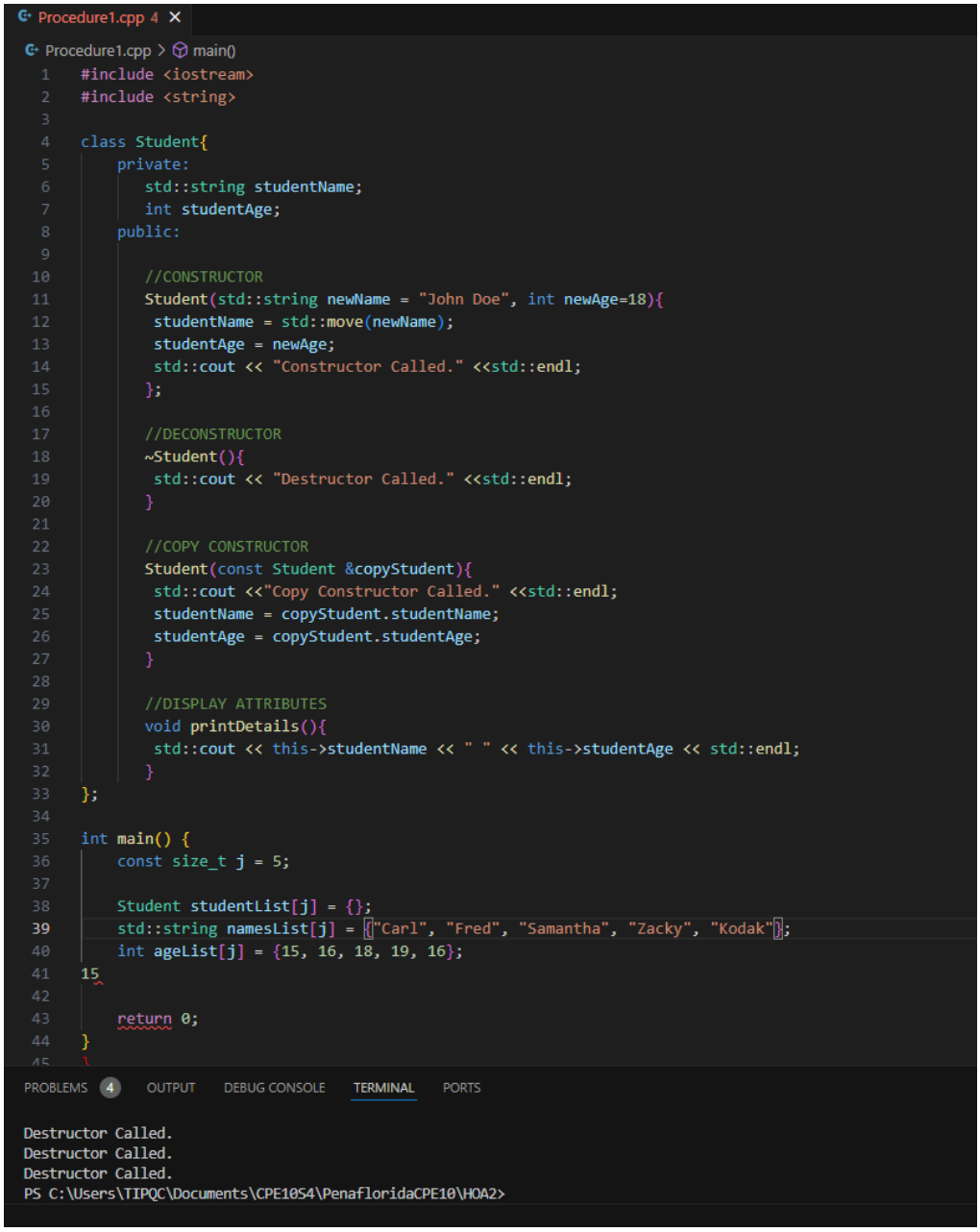
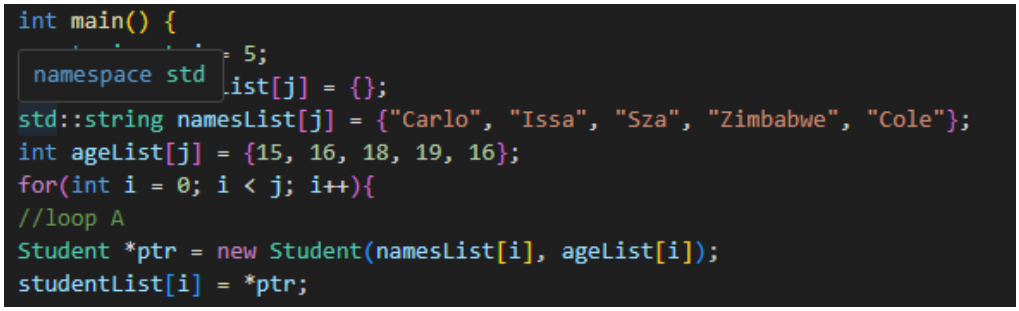
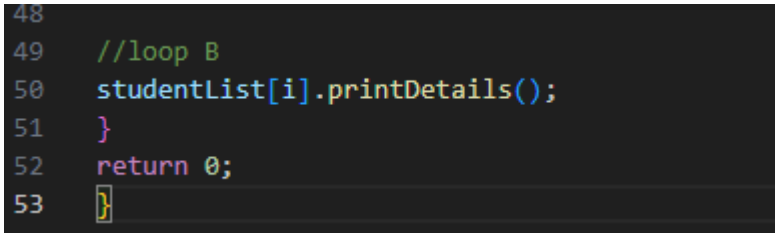
Screenshot	
Observation	This run creates student1 (constructor), student2 (copy constructor from student1), and student3 (default constructor). student3 = student2 triggers the assignment operator, not the copy constructor, so no “Copy Constructor Called.” message appears for that line. All 3 objects go out of scope at the end of main(), producing the 3 “Destructor Called.” lines seen at the bottom of the terminal

Table 2-1. Initial Driver Program

Screenshot	
Observation	In this modified program using arrays instead of individual objects — studentList[j], namesList[j], and ageList[j] are declared to hold 5 students, but the loop that populates and processes them (Loop A/Loop B) isn't shown in this snippet. The terminal only shows 3

	"Destructor Called." lines, which is unexpected — with j = 5, there should be 5 destructor calls if all list elements were properly constructed and destroyed, suggesting the array may not have been fully populated before the program exited, or some output scrolled out of view.

Table 2-2. Modified Driver Program with Student Lists

Loop A	
Observation	Even though studentList = *ptr looks like it should call the copy constructor, it doesn't — the copy constructor is never triggered. Since studentList was already initialized with {} (which calls the default constructor 5 times), assigning *ptr to studentList[uses the default copy assignment operator instead, which doesn't print anything. That's why the output shows exactly 10 "Constructor Called." lines (5 from {} init + 5 from new) and zero "Copy Constructor Called." lines. Also, ptr is never deleted, so there's a memory leak for each iteration.
Loop B	
Observation	Loop B correctly outputs each student's name and age in order (Carlo 15, Issa 16, Sza 18, Zimbabwe 19, Cole 16), confirming the data was properly copied into studentList[i] even without an explicit copy constructor call — the compiler-generated assignment operator handled the memberwise copy correctly

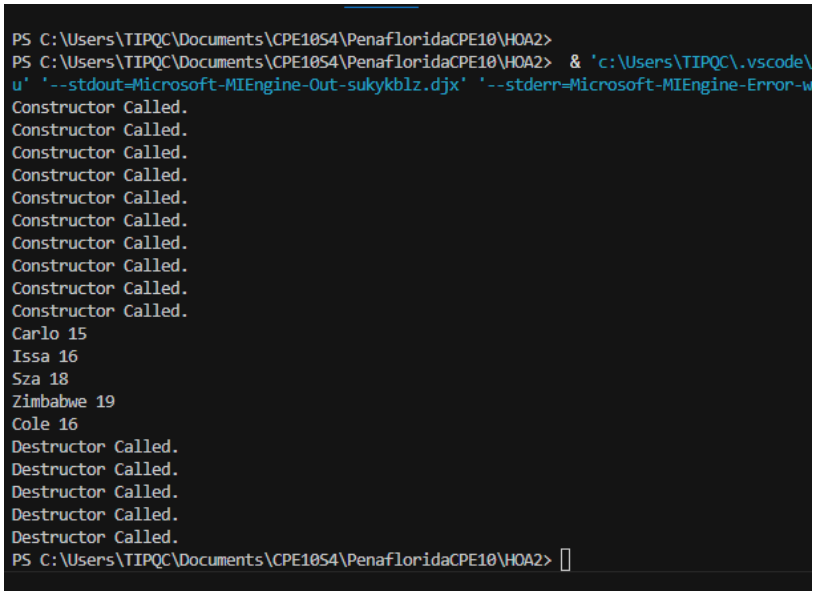
Output	
Observation	10 constructor is called followed by correct names and ages then afterwards 5 destructor was called and get destroyed when the array go out of scope.

Table 2-3. Final Driver Program

Modifications	
Observation	

Table 2-4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For general questions like installing the c++ extension and how to set it up on my laptop**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Only help me to install the c++ extensions and missing files that is needed for this task to run on my vscode**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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